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Data Types in C

*What's in a name? That which we call a rose,
By any other name would smell as sweet.*

William Shakespeare, Romeo and Juliet,
Act II, Scene II

- What is a “variable”?
- Data Type in C
- Numbers
 - Floating point data types
 - Integer data types
- Combining different types
 - Cast
- Visibility & Lifetime

SUMMARY



- Something that is able to “vary”
- Definition:
 - **Memory space** with a **name** where it is possible to **store data** during program execution
- Memory space $\rightarrow$ address & size
 - Automatically managed by high level programming languages

- Content or value of a variable → right value or rvalue
- Memory address → left value or lvalue
- Size

```
int a = 173;
```

- Rules
 - Only letters, digits, or underscore “_”
 - It must begin with a letter (or “_”)
 - Names are case sensitive
 - Reserved words can not be used

- Recommended style:
 - Select small caps for variable names
 - Use meaningful names
 - When the name is composed by multiple words use uppercase letters or underscore to emphasize that

```
int  sum           = 173;  
int  students_number = 144;  
int  ClassId       = 10;
```

How to define a variable

```
// variables are defined as
```

```
// <type> <variable_name>;
```

```
int a;
```

```
    // I am defining a “int” variable named ‘b’
```

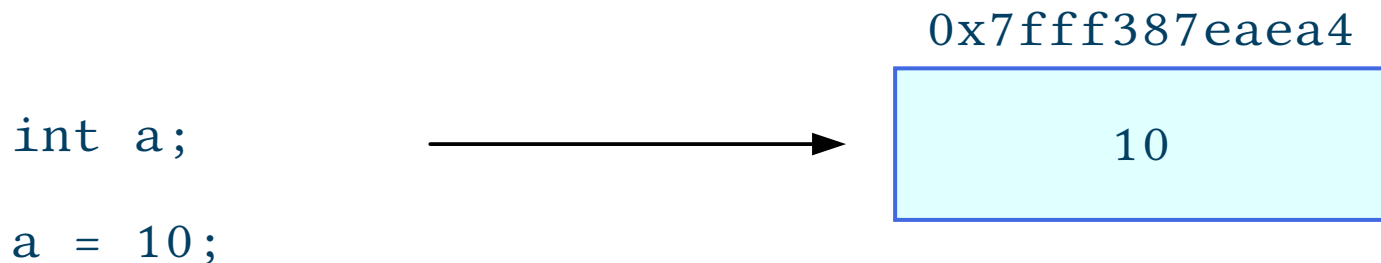
```
    // in such a case the variable is NOT initialized
```

```
    // this means that ‘a’ contains a RANDOM number
```

```
int b=17;
```

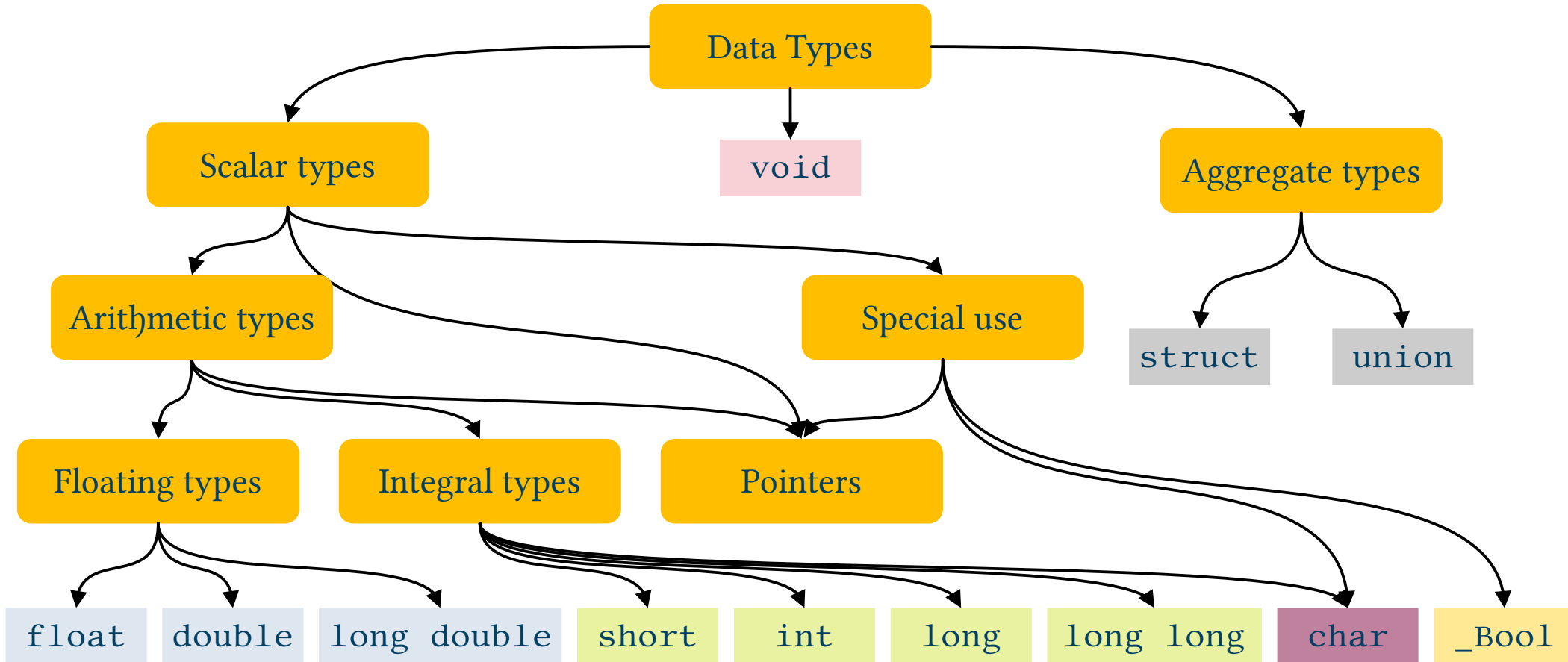
```
    // I am defining AND initializing a “int” variable named ‘b’
```

Defining & initing a variable

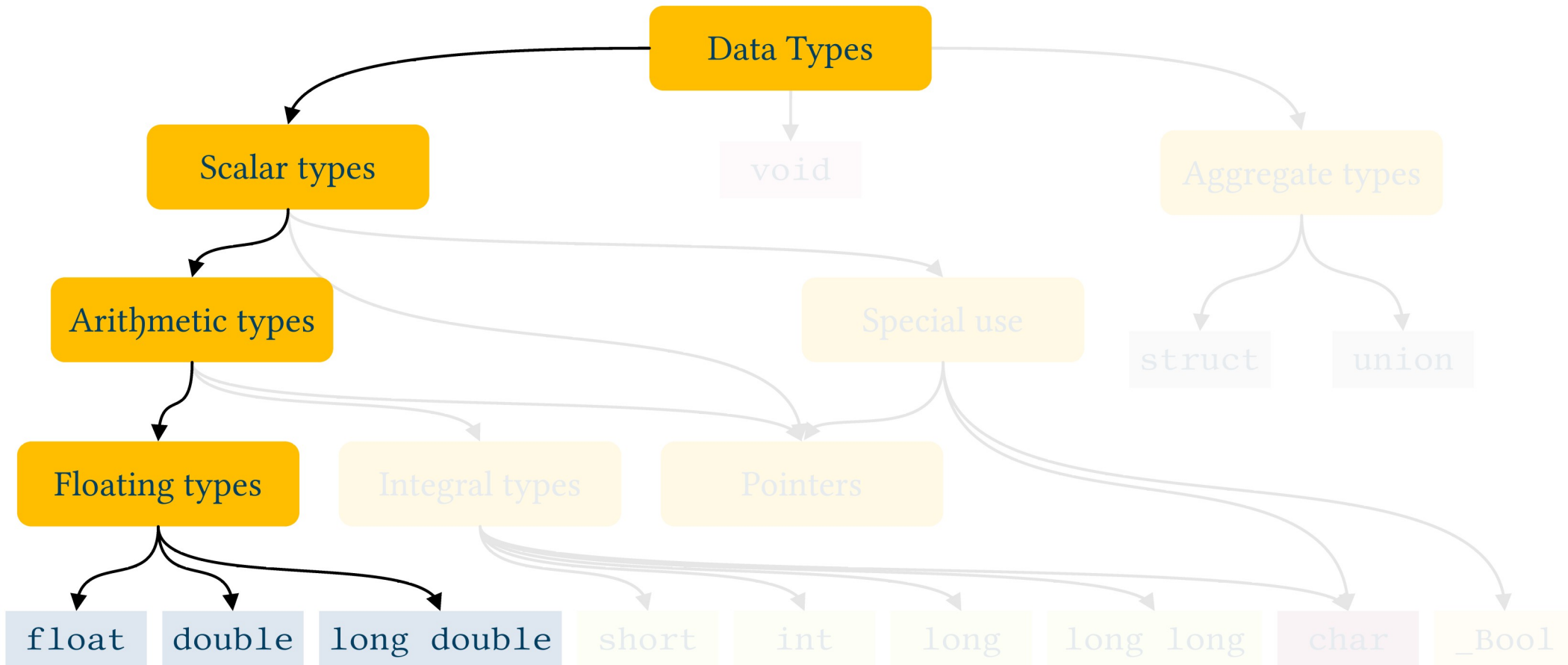


- Definition of a variable does not initialize it
 - Random value
 - Do not rely on it being 0
- Always init variables before using them!

C99 Hierarchy of Data Types (simplified)

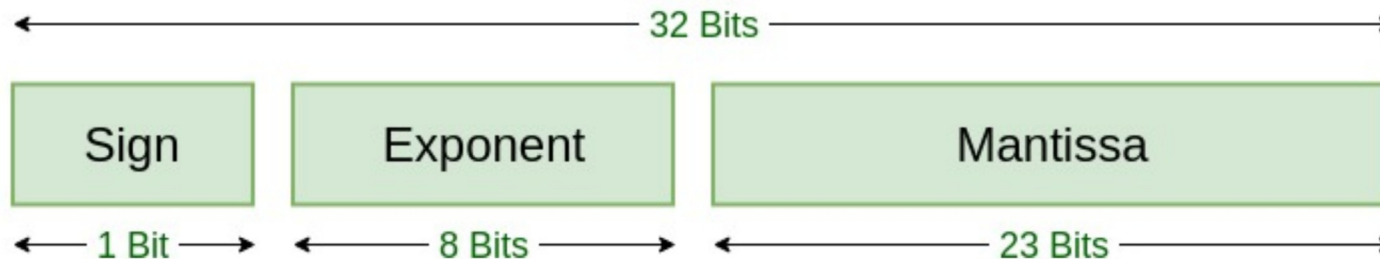
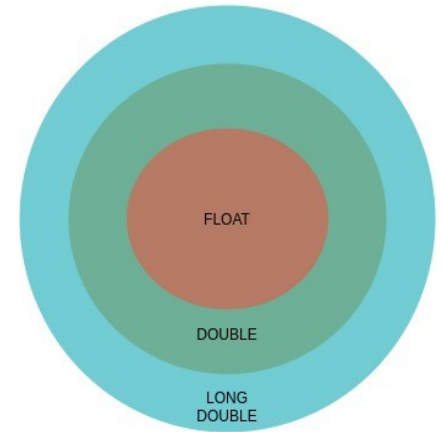


Floating point types



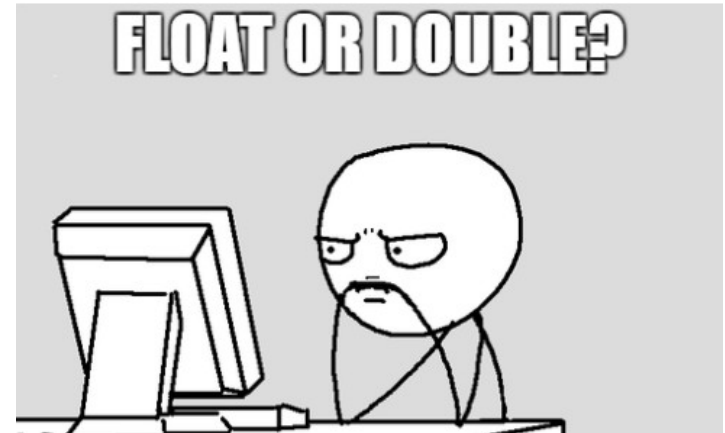
- They can hold very large numbers or fractions
 - Positive or negative numbers
 - With or without the fractional part
 - Remember approximation
- Why we have approximation?
 - Limited size for the fractional part
 - Min & Max values limit

- 3 possibilities
 - float → single precision
 - double → “double” precision
 - long double → extended precision
- Main difference?
 - Number of bytes used to store the data



Float vs Double vs Long Double

- Which type should I use?
- As always it is a compromise amongst:
 - Precision
 - Execution speed
 - Memory requirements



- The C standard recommends the IEEE 754 format?
 - Actually not, but this is not important
- The C standard says anything about the size of different types?
 - No
- It is up to the compiler!
 - Often, there is not difference between float or double types

- We have a tool to understand the size of a variable
- sizeof(<data type>) or sizeof(<variable name>)
 - It returns the number of bytes used to store that data
- A specific prefix ('z') for a correct format specifier
 - %zd C99
 - %ld C89 – use this one...

- How to print or read them?

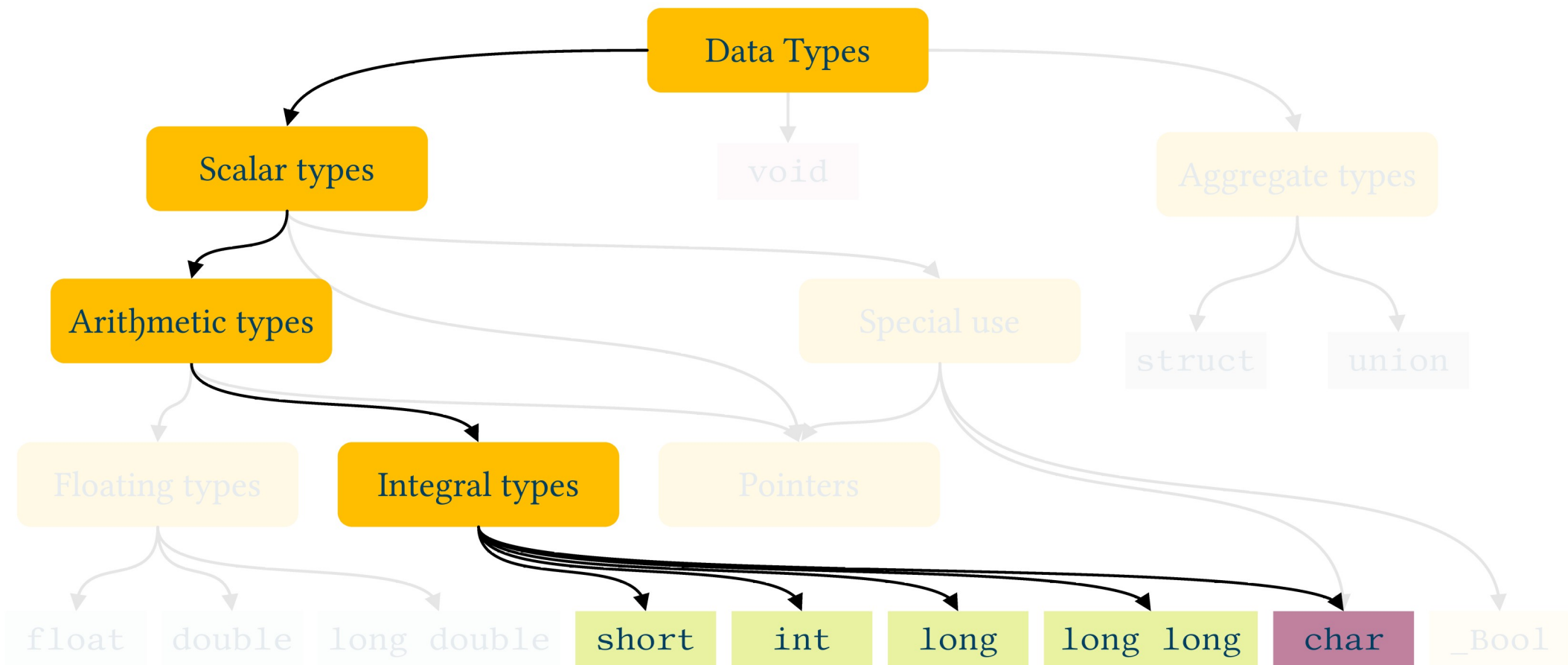
- Formats

- %f → decimal notation 123.456000
 - %e → scientific notation 1.234560e+02
 - %E → scientific notation 1.234560E+02
 - %g → shortest between %f and %e 123.456

- Suffissi per tipo

- “none” → float
 - l → double (only mandatory for input)
 - L → long double

- Specific values
- $\pm\text{inf}$ $\rightarrow$ Infinite
 - Usually as a result of math operations
 - i.e. division by zero
- NaN $\rightarrow$ not a number
 - Probably some memory error
- Specific functions
 - `isnan()`, `isinf()`...



- 4+1 types:
 - char → seldom used for “numbers”
 - short int
 - int → probably the most used data type
 - long int
 - long long int
- The difference is, again, the size of memory used to store them
 - Given n bits I can store non negative numbers in the $0 - 2^n - 1$ interval

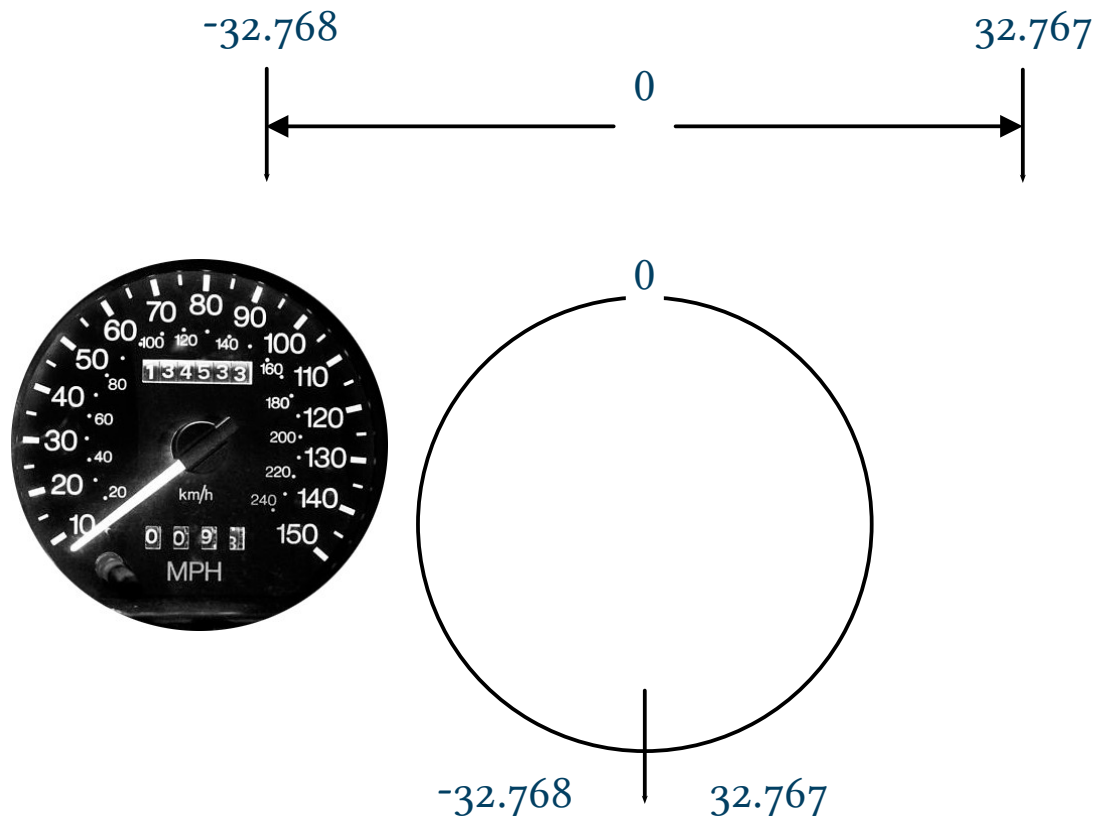
- All integer types can deal with negative numbers (“signed” types)
 - The range of numbers I can store is then not $[0, 2^n-1]$ but $[-2^{n-1}, 2^{n-1}-1]$
- By default all signed types but
 - char – compiler dependent
- When I use them I can change the default behavior
 - unsigned
 - (signed)
- This brings the number of possible integer types to 10!

Integer Types (typical example for 64 bit)

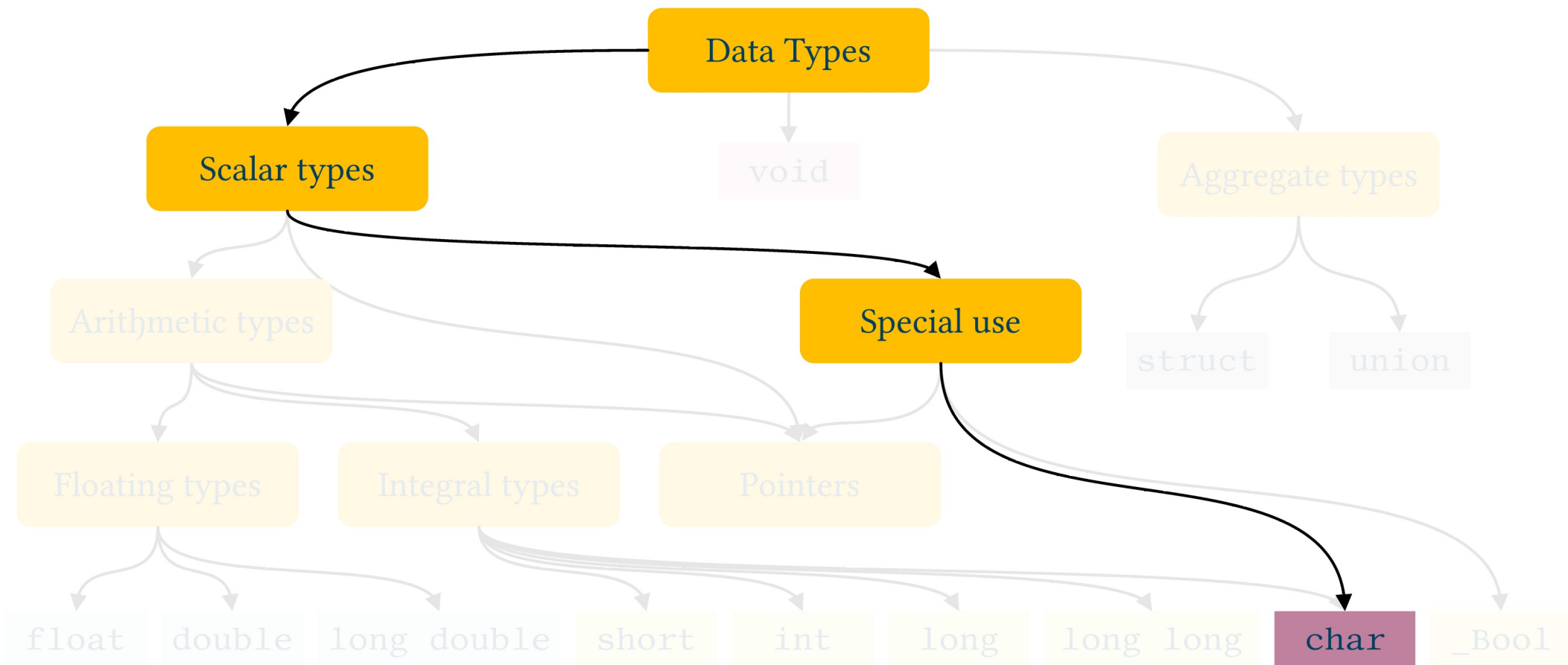
Type	Byte	Smallest Value	Largest Value
signed char	1	-128	127
unsigned char	1	0	255
short	2	-32.768	32.767
unsigned short	2	0	65.535
int	4	-2.147.483.648	2.147.483.647
unsigned int	4	0	4.294.967.295
long	8	-9.223.372.036.854.775.808	9.223.372.036.854.775.807
unsigned long	8	0	18.446.744.073.709.551.615
long long	8	-9.223.372.036.854.775.808	9.223.372.036.854.775.807
unsigned long long	8	0	18.446.744.073.709.551.615

- How we can read or print them?
 - Formats
 - `%d` → decimal notation -45
 - `%x` → hexadecimal notation fffffd3
 - `%X` → hexadecimal notation FFFFFFFD3
 - `%u` → decimal notation for unsigned types 4294967251
 - Suffixes
 - “none” → int
 - `hh` → char
 - `h` → short
 - `l` → long
 - `ll` → long long

- When an operation produces a result too big (or too small) for the given type
- No execution error!
 - Usually clamping



Char Type



- It is used to store numbers!
 - 0–255
 - -128–127
- But typically only used to store ASCII codes
- Format specifier
 - %c
- Constants:
 - ~~Direct use of magic numbers, i.e. 65~~
 - Use of correspondent symbol 'A'



Do not
even think
to do that!

The char type & scanf()

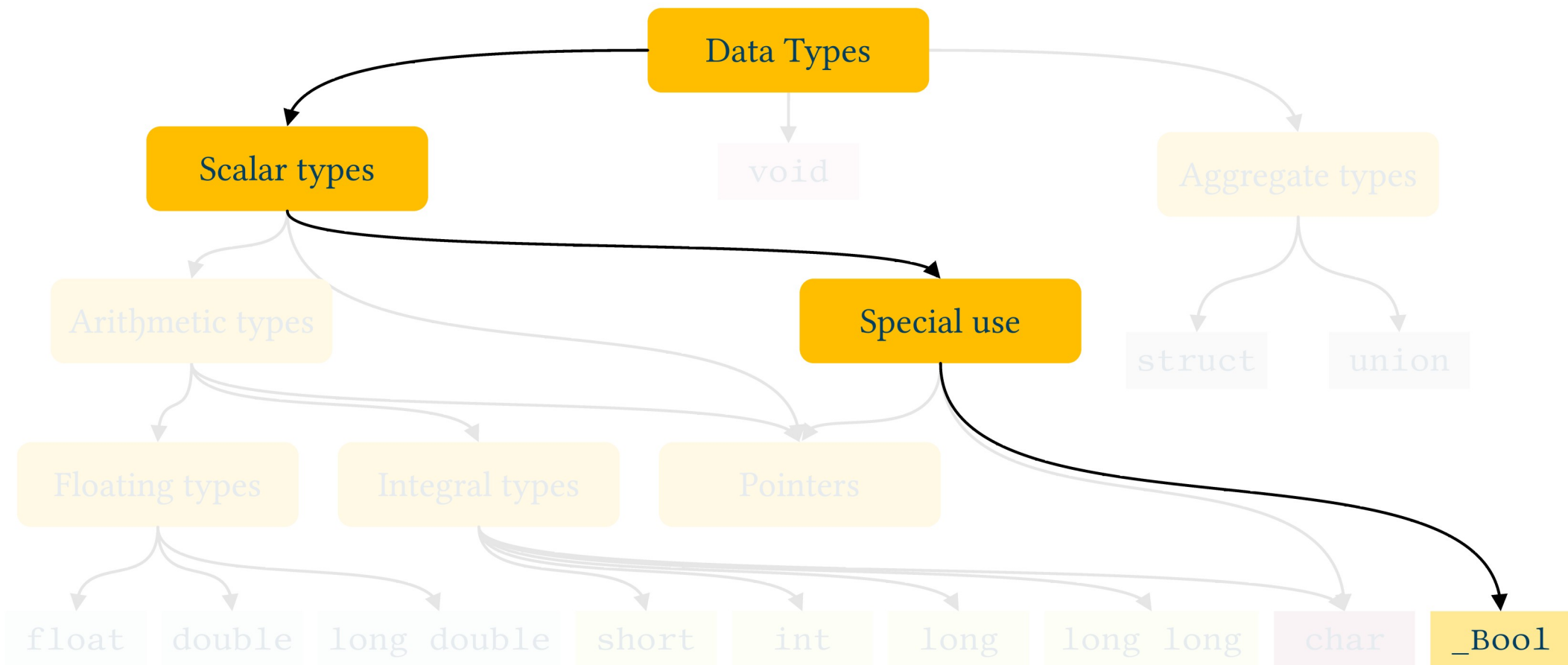
- Reading the char and derived types using scanf() can be a little tricky
- When I type on the keyboard all characters are fed as input to the program
 - All means all. Namely, also the enter key...
 - All those “key presses” are initially stored in a input buffer



```
scanf("%c", &x); // I also read spaces and newlines
```

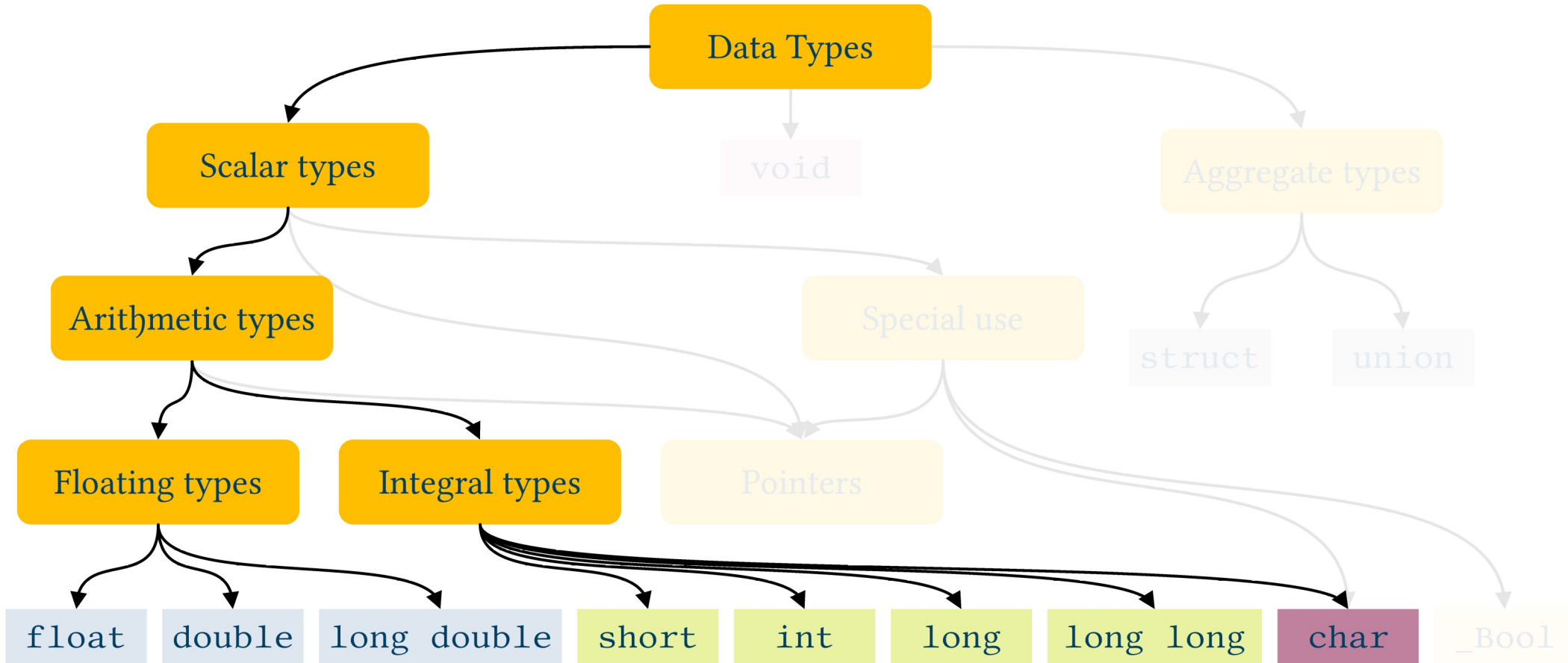
```
scanf(" %c", &x); // Ignore leading spaces or newlines
```

Boolean Type



- C99 (and later) specific
- A integer data type with the aim to only store “true” or “false”
 - $0 \rightarrow \text{false}$
 - $1 \rightarrow \text{true}$
- This type can then features 1 or 0 values only

“Arithmetic” issues

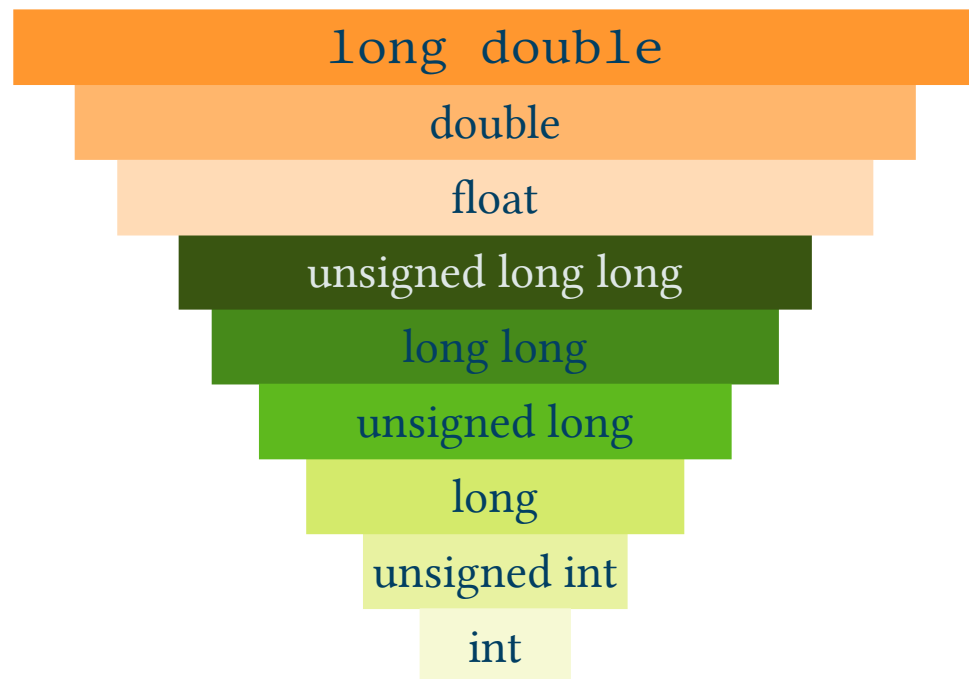


- 3 floating point types and 8–10 integer ones
- Can I combine them in operations?
 - Yes
- What is the C behavior when I mix different kind of data types?
- Conversions
 - Assignment
 - Implicit or automatic conversions

- When I operate an assignement (=)
 - The right operand (rvalue) is “casted” to the type I have on the left operand (lvalue)
- Issues
 - I can lost information i.e.: float $\rightarrow$ int
 - Wrong result i.e.: long long $\rightarrow$ short

- It occurs when we combine different types in expressions
- Binary operands needs same type operators
 - Arithmetic: $*$, $/$, $\%$, $+$, $-$; Comparisons: $<$, $>$, $\leq$, $\geq$, $==$, $!=$; Bitwise: $\&$, $\wedge$, $|$,
- i.e.: $a+b$ when a is an int and b is a float which kind of type gives as a result?
 - Spoiler: float
- Conversion rules
- High probability of bugs!

- Some types are converted to int or unsigned int
 - Even when we have same type operands
 - `_Bool`, `char` e `short` $\rightarrow$ int
 - Or unsigned int
- After this $\rightarrow$ conversion pyramid



- I select the conversion
 - This lead to avoid errors
- Syntax:

(type) expression

- Can I use a variable everywhere in my code?
 - No!
- Visibility field or scope
 - Part of the code where a variable can be used
- 3 cases:
 - Program (global variables)
 - File
 - Block (local variables)



- Defined outside of every code block
 - {}
- I can modify them everywhere
 - Data are NOT protected
 - Debug becomes more difficult
 - Uncertain
- Your final score will be, FOR SURE, affected
- One single benefit → initialization to 0



- Defined
 - Inside a code block i.e. inside `{}`
 - In a `for(;;)`
 - As function arguments
- I can use them only inside the specif block where they are defined
- Automatic duration
 - Created when the execution flow meets that code block
 - Destroyed when that code block is over

- Local Variables
 - Let code blocks or functions to be self-sufficient
 - Easier to manage or to understand the code
 - Modularity
 - Less portability issues
- Global Variables
 - I have to track all the code parts where they are used
 - Require to coordinate team work
 - Memory is always used
 - Naming issues

- A variable is not always forever
- Static lifetime
 - Global Variables
 - Memory reserved when the execution begins and never freed
- Automatic lifetime
 - Local variables
 - Local variables are created when code execution reaches that block
 - And destroyed when the execution is over
 - Can I change this behavior? Yes → `static`



- We can need to use constant values
- Quick and dirt approach:
 - Put in my code a lot of “magic numbers”
 - This lead to a code really difficult to modify or maintain
- Good approach:
 - Define constant
 - Parametric code, very easy to modify or improve

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How to define a constant value in C

- Three different approaches
- `#define`
 - i.e.: `#define N_MAX_STUDENTS (100)`
- `const` prefix
 - I define and init a variable whose initial value can not be modified
 - In such a case the use of a global variable makes a sense
 - i.e.: `const int N_MAX_STUDENTS=100;`



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KEEP
CALM
IT'S
QUESTION
TIME

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William Shakespeare, Romeo and Juliet,
Act II, Scene II